Tutorial - 1

1. Suppose that (sets)

$A = \{x : x \in N \text{ and } x \text{ is even}\}$
- set of even natural numbers, $A = \{2, 4, 6, 8, 10, \ldots\}$

$B = \{x : x \in N \text{ and } x \text{ is prime}\}$,
- set of prime numbers, $B = \{2, 3, 5, 7, 11, \ldots\}$

$C = \{x : x \in N \text{ and } x \text{ is a multiple of } 5\}$
- multiple of 5, $C = \{5, 10, 15, 20, 25, \ldots\}$

(a) $A \cap B = \{2\}$
(b) $B \cap C = \{5\}$
(c) $A \cup B = \{2, 3, 4, 5, 6, 7, 8, 10, 11, \ldots\}$
(d) $A \cap (B \cup C) = \{2, 10, 20, 30\}$

2) Matrix Operation

1. a) 3x3 matrix with ones on diagonal

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

b) 3x4 matrix with ones on diagonal

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

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c) 4×3 matrix with ones on diagonal

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

2) Compute AD and DA.

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 4 \end{bmatrix} \quad D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$AD = \begin{bmatrix} 1\times2+1\times0+1\times0 & 1\times0+1\times3+1\times0 & 1\times0+1\times0+1\times4 \\ 1\times2+2\times0+3\times0 & 1\times0+2\times3+3\times0 & 1\times0+2\times0+3\times4 \\ 1\times2+3\times0+4\times0 & 1\times0+3\times3+4\times0 & 1\times0+3\times0+4\times4 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 3 & 4 \\ 2 & 6 & 12 \\ 2 & 9 & 6 \end{bmatrix}$$

$$DA = \begin{bmatrix} 2\times1+0\times1+0\times1 & 2\times1+0\times2+0\times3 & 2\times1+0\times3+0\times4 \\ 0\times1+3\times1+0\times1 & 0\times1+3\times2+0\times3 & 0\times1+3\times3+0\times4 \\ 0\times1+0\times1+4\times1 & 0\times1+0\times2+4\times3 & 0\times1+0\times3+4\times4 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 2 & 2 \\ 3 & 6 & 9 \\ 4 & 12 & 16 \end{bmatrix}$$

$\therefore AD \neq DA$

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3) $A=\begin{bmatrix}1 & 2\\ 2 & 4\end{bmatrix}$, $B=\begin{bmatrix}2 & 1\\ 1 & 3\end{bmatrix}$, $C=\begin{bmatrix}4 & 3\\ 0 & 2\end{bmatrix}$, verify that $AB=AC$

$$AB=\begin{bmatrix}1 & 2\\ 2 & 4\end{bmatrix}\begin{bmatrix}2 & 1\\ 1 & 3\end{bmatrix}=\begin{bmatrix}1\times2+2\times1 & 1\times1+2\times3\\ 2\times2+4\times1 & 2\times1+4\times3\end{bmatrix}=\begin{bmatrix}4 & 7\\ 8 & 14\end{bmatrix}$$

$$AC=\begin{bmatrix}1 & 2\\ 2 & 4\end{bmatrix}\begin{bmatrix}4 & 3\\ 0 & 2\end{bmatrix}=\begin{bmatrix}1\times4+2\times0 & 1\times3+2\times2\\ 2\times4+4\times0 & 2\times3+4\times2\end{bmatrix}=\begin{bmatrix}4 & 7\\ 8 & 14\end{bmatrix}$$

$\therefore AB=AC$

4) Prove $(AB)^T=B^TA^T$

Here,

A be $m\times n$ matrix, B be $n\times p$ matrix

product $C=AB$ is an $m\times p$ matrix

element of $C_{ij}=\sum_{k=1}^{n} a_{ik}\,b_{kj}$

Transpose of AB,

$(AB)^T_{ij}=AB_{ji}$

Substituting the formula of matrix multiplication

$$(AB)_{ji}=\sum_{k=1}^{n} a_{jk}.b_{ki}$$

Now,

$A^T_{kj}=a_{jk}$

$B^T_{ik}=b_{kj}$

After multiplying and substituting, $\sum_{k=1}^{n} a_{jk}b_{ki}$

Comparing, $(AB)^T_{ij}=\sum a_{jk}\,b_{ki}$, $(B^TA^T)_{ij}=\sum a_{jk}\,b_{ki}$

$\therefore (AB)^T=B^TA^T$ //

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5) Let $A = \begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix}$, compute A^TA.

Here, $A^T = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$

$$A^T \cdot A = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} \begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix} = \begin{bmatrix} a^2+b^2+c^2 & ad+be+cf \\ ad+be+cf & d^2+e^2+f^2 \end{bmatrix}$$

∴ A matrix is symmetric if $M = M^T$, element (1,2) and (2,1) have $ad+be+cf$.

So, $A^TA = (A^TA)^T$ is symmetric.

6) Find inverse of

$$A = \begin{bmatrix} 5 & 6 \\ 4 & 5 \end{bmatrix}$$

$$|A| = \begin{bmatrix} 5 & 6 \\ 4 & 5 \end{bmatrix} = 5\times5 - 4\times6 = 1$$

$$A^{-1} = \begin{bmatrix} 5 & -6 \\ -4 & 5 \end{bmatrix}$$

$$B = \begin{bmatrix} 6 & 4 \\ 3 & 3 \end{bmatrix} = 6\times3 - 3\times4 = 6$$

$$B^{-1} = \begin{bmatrix} 1/2 & -2/3 \\ -1/2 & 1 \end{bmatrix}$$

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7) Verify A and B are inverse

$$A=\begin{bmatrix}4 & 1\\ 3 & 1\end{bmatrix},\quad B=\begin{bmatrix}1 & -1\\ -3 & 4\end{bmatrix}$$

$$AB=\begin{bmatrix}4 & 1\\ 3 & 1\end{bmatrix}\begin{bmatrix}1 & -1\\ -3 & 4\end{bmatrix}=\begin{bmatrix}4\times1-1\times(3) & 4\times(-1)+1\times4\\ 3\times1-1\times(-3) & 3\times(-1)+1\times4\end{bmatrix}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$

$$BA=\begin{bmatrix}1 & -1\\ -3 & 4\end{bmatrix}\begin{bmatrix}4 & 1\\ 3 & 1\end{bmatrix}=\begin{bmatrix}1\times4-1\times3 & 1\times(-1)\times1\\ -3\times4+4\times3 & -3\times1+4\times1\end{bmatrix}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$

$\therefore AB=BA$ //

8) Verify given matrix are inverses of each other.

$$AB=\begin{bmatrix}1 & -1 & 0\\ 1 & 0 & -1\\ 2 & 3 & 4\end{bmatrix}\begin{bmatrix}3 & -4 & 1\\ 2 & -4 & 1\\ 3 & -5 & 1\end{bmatrix}=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$$

$$BA=\begin{bmatrix}3 & -4 & 1\\ 2 & -4 & 1\\ 3 & -5 & 1\end{bmatrix}\begin{bmatrix}1 & -1 & 0\\ 1 & 0 & -1\\ 2 & 3 & -4\end{bmatrix}=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$$

$\therefore AB=BA$

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3.4 Solving Linear equation using matrix method,

1.a) $\begin{cases} x+2y=4 \\ 3x-5y=1 \end{cases}$

$A=\begin{bmatrix} 1 & 2 \\ 3 & -5 \end{bmatrix}, B=\begin{bmatrix} 4 \\ 1 \end{bmatrix}, C=\begin{bmatrix} x \\ y \end{bmatrix}$

$|A|=\begin{bmatrix} 1 & 2 \\ 3 & -5 \end{bmatrix} = 1\times(-5)-3\times2=-11$

$A^{-1}=\frac{1}{-11}\begin{bmatrix} -5 & -2 \\ -3 & 1 \end{bmatrix}$

6. $X=A^{-1}B$

$=\frac{1}{-11}\begin{bmatrix} -5 & -2 \\ -3 & 1 \end{bmatrix}\begin{bmatrix} 4 \\ 1 \end{bmatrix}$

$=\frac{1}{-11}\begin{bmatrix} -22 \\ -11 \end{bmatrix}=\begin{bmatrix} 2 \\ 1 \end{bmatrix}$

$\therefore x=2, y=1$

b) $\begin{cases} 5x+y=13 \\ 3x+2y=5 \end{cases}$

$|A|=\begin{bmatrix} 5 & 1 \\ 2 & 3 \end{bmatrix} = 5\times2-3\times1=7$

$A^{-1}=\frac{1}{7}\begin{bmatrix} 2 & -1 \\ -3 & 5 \end{bmatrix}$

$Y=A^{-1}B=\frac{1}{7}\begin{bmatrix} 2 & -1 \\ -3 & 5 \end{bmatrix}\begin{bmatrix} 13 \\ 5 \end{bmatrix}$

$=\frac{1}{7}\begin{bmatrix} 2\times13-1\times5 \\ -3\times13+5\times5 \end{bmatrix}=\begin{bmatrix} 21 \\ -14 \end{bmatrix}$

$\therefore x=3, y=-2$

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c) $\begin{cases} 3x+2y=-2 \\ x+4y=6 \end{cases}$

$|A| = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix} = 3\times4 - 1\times2 = 10$

$A^{-1} = \frac{1}{10}\begin{bmatrix} 4 & -2 \\ -1 & 3 \end{bmatrix}$

$X = A^{-1}B = \frac{1}{10}\begin{bmatrix} 4 & -2 \\ -1 & 3 \end{bmatrix}\begin{bmatrix} -2 \\ 6 \end{bmatrix} = \begin{bmatrix} 4\times(-2) - 2\times6 \\ -1\times(-2) + 3\times6 \end{bmatrix} = \begin{bmatrix} -20 \\ 20 \end{bmatrix}$

$= \frac{1}{10}\begin{bmatrix} -20 \\ 20 \end{bmatrix}$, $\therefore \begin{bmatrix} -2 \\ 2 \end{bmatrix}$, $x=-2, y=2$ //

d) $\begin{cases} 2x+3y=6 \\ x-y=\frac{1}{2} \end{cases}$

$|A| = \begin{bmatrix} 2 & 3 \\ 1 & -1 \end{bmatrix} = 2\times(-1) - 1\times3 = -2-3 = -5$

$A^{-1} = \frac{1}{-5}\begin{bmatrix} -1 & -3 \\ -1 & 2 \end{bmatrix}$

$X = A^{-1}B = \frac{1}{-5}\begin{bmatrix} -1 & -3 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 6 \\ 1/2 \end{bmatrix}$

$= \frac{-1}{5}\begin{bmatrix} -1\times6 - 3\times1/2 \\ -1\times6 + 2/2 \end{bmatrix}$

$= \begin{bmatrix} 5/7 \\ 1/7 \end{bmatrix}$

$\therefore x = \frac{5}{7}, y = \frac{1}{7}$ //

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e) $\begin{cases} 4x+3y=11 \\ x-3y=-1 \end{cases}$

$|A| = \begin{vmatrix} 4 & 3 \\ 1 & -3 \end{vmatrix} = -12-3 = -15$

$A^{-1} = \frac{1}{-15}\begin{bmatrix} -3 & -3 \\ -1 & 4 \end{bmatrix}$

$X = A^{-1}B = \frac{1}{-15}\begin{bmatrix} -3 & -3 \\ -1 & 4 \end{bmatrix}\begin{bmatrix} 11 \\ -1 \end{bmatrix}$

$= \frac{1}{-15}\begin{bmatrix} -3\times11+(-3)\times(-1) \\ -1\times11+4\times(-1) \end{bmatrix} = \begin{bmatrix} -33+3 \\ -11-4 \end{bmatrix} = \begin{bmatrix} -30 \\ -15 \end{bmatrix}$

$= \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $x=2, y=1$ //

f) $\begin{cases} 2x+3y=8 \\ 4x+6y=20 \end{cases}$

$|A| = \begin{vmatrix} 2 & 3 \\ 4 & 6 \end{vmatrix} = 12-12 = 0$, no solution

2.a) $\begin{cases} x+y-z=2 \\ x+z=7 \\ 2x+y+z=13 \end{cases}$

Here,

$x = 7-z$

$7-z+y-z=2$

$7-2z+y=2$

$5+y=2z$

$\therefore z = \frac{5+y}{2}$

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$2(7-z)+y+\frac{5+y}{2}=13$

$14-2z+y+\frac{5+y}{2}=13$

$28-4z+2y+5+y=26$

$7+3y-4z=0$

$7+3y-\frac{4}{2}\left(\frac{5+y}{2}\right)=0$

$7+3y-10-2y=0$

$-3+y=0$

$\therefore y=3,\ z=\frac{5+3}{2}=\frac{8}{2}=4,\ x=7-2=7-4=3$

$\therefore x=3, y=3, z=4$ //

b) $\begin{cases} x+y-z=2 \\ 3x+y=7 \\ x+y+2z=3 \end{cases}$

Here,

$3x+y=7 \quad \ldots(i)$

$y=7-3x$

(i) $x+y-z=2$

Substitute $y=7-3x$

$x+(7-3x)-z=2$

$7-2x-z=2$

$5-2x=z$

$z=5-2x$

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Substitute in eqn (ii)

$y = 7 - 3x$ and $z = 5 - 2x$

$x + (7 - 3x) + 2(5 - 2x) = 3$

$x + 7 - 3x + 10 - 4x = 3$

$17 - 6x = 3$

$6x = 14$

$x = \frac{14}{6} = \frac{7}{3}$

$y = 7 - 3x$

$y = 7 - 3\left(\frac{7}{3}\right)$

$y = 7 - 7 = 0$

$z = 5 - 2x$

$z = 5 - 2\left(\frac{7}{3}\right) = \frac{1}{3}$

$\therefore x = \frac{7}{3}, y = 0, z = \frac{1}{3}$

c) $\begin{cases} -x - 2y + z = 1 \\ 2x + 3y = 2 \\ y - 2z = 0 \end{cases}$

(iii) $y - 2z = 0$

$y = 2z$

(ii) $2x + 3y = 2$

Substitute $y = 2z$

$2x + 3(2z) = 2$

$2x + 6z = 2$

$x + 3z = 1$

$x = 1 - 3z$

(i) $-x - 2y + z = 1$

$x = 1 - 3z$ and $y = 2z$

$-(1 - 3z) - 2(2z) + z = 1$

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$-1+3z-4z+z=1$

$-1+0=1$

$-1=1$, no solution

d) $\begin{cases} x+4y-z=4 \\ 2x+5y+8z=15 \\ x+3y-3z=1 \end{cases}$

express x from (i) $x+4y-z=4$

$x=4-4y+z$

(ii) $x+3y-3z=1$

substitute $x=4-4y+z$

$(4-4y+z)+3y-3z=1$

$4-y-2z=1$

$y+2z=3$

$y=3-2z$

substitute x and y into eqn (ii)

$2x+5y+8z=15$

substitute $x=4-4y+z$

$2(4-4y+z)+5y+8z=15$

$8-8y+2z+5y+8z=15$

$8-3y+10z=15$

$-3y+10z=7$

$y=3-2z$

$y=3-2(1)$

$y=1$

Substitute $y=3-2z$

$-3(3-2z)+10z=7$

$-9+6z+10z=7$

$16z=16$

$z=1$

$x=4-4y+z$

$x=4-4(1)+1$

$=0(1)+1$

$=0+1$

$=1$

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e) $\begin{cases} 5x+3y+9z=-1 \\ -2x+3y-z=-2 \\ -x-4y+5z=1 \end{cases}$

Here,

express x from equation (iii) $-x-4y+5z=1$

multiply by -1 : $x-4y-5z=-1$

$x=-1-4y+5z$

Substitute $x=-1-4y+5z$

$-2(-1-4y+5z)+3y-z=-2$

$2+8y-10z+3y-z=-2$

$2+11y-11z=2$

$11y-11z=-4$

$y-z=\frac{-4}{11}$

$y=z-\frac{4}{11}$

Sub eqn (ii) from (i)

~~$(5x+3y+9z)-(2x+3y-z)=$~~

$5x+3y+9z+2x-3y+z$

$=7x+10z$:

~~$-1(-2)=1$~~

~~$-36-$~~ $=-1-(-2)=1$

$=7x+10z=1$ //

Adding eqn (ii) and (iii)

$(2x+3y-z)+(-x-4y+5z)$

$-3x-y+4z=-1$

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$7x + 10z = 1$

$-3x - y + 4z = -1$

solving eq(i) together,

$z = \frac{24}{187}$ $\quad x = \frac{61}{187}$

$y = \frac{92}{187}$

f) $\begin{cases} 6x - 5y + 2z = 3 \\ 2x + y - 4z = 5 \\ 3x - 3y + z = 1 \end{cases}$

Proof,

multiply eqn(ii) by 3

$6x + 3y - 12z = 15$

Subtract eqn(i)

$8y - 14z = 12$

multiply (iii) by 2

$6x - 6y + 2z = -2$

Subtract eqn(i), $y = 5$

Using eqn(ii)

$2x + 5 - 4z = 5$

$2x - 4z = 0$

$x = 2z$

Now,

$3(2z) - 3 \times 5 + z = -1$

$6z - 15 + z = -1$

$7z = 14$

$z = \frac{14}{7} z = 2$

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4.1) Find $y'(x)$

1) $f(x) = x + \sqrt{x}$

$f'(x) = \frac{d}{dx}\{x + \sqrt{x}\}$

$= \frac{d}{dx} + \frac{d\sqrt{x}}{dx}$

$= 1 + \frac{d}{dx}\{x^{1/2}\}$

$= 1 + (1/2)x^{(1/2)-1}$

$= 1 + (1/2)x^{-1/2}$

$= 1 + \frac{1}{2\sqrt{x}}$

2) $f(x) = 2 + 1/x + 1/x^2$

or $f(x) = 2 + 1x^{-1} + x^{-2}$

or, $0 - x^{-2} - 2x^{-3}$

or, $\frac{-1}{x^2} - \frac{2}{x^3}$

3) $f(x) = \sqrt{x}\cos(x)$

$u = \sqrt{x} = x^{1/2}$, $v = \cos x$

$\frac{du}{dx} = \frac{d}{dx}(x^{1/2}) = \frac{1}{2\sqrt{x}}$

$\frac{dv}{dx} = \frac{d}{dx}(\cos x) = \sin x$

$f'(x) = \frac{1}{2\sqrt{x}}\cos x - \sqrt{x}\sin x$

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4) $f(x) = x^3 e^x$

let $u = x^3$, $v = e^x$

$\frac{du}{dx} = \frac{dx^3}{dx} = 3x^2$

$f'(x) = 3x^2 e^x - x^3 \cdot x^3 e^x (3x^2 - x^3)$

5) $\frac{x^3}{e^x}$

$f(x) = x^3 e^{-x}$

$u = x^3$, $v = e^{-x}$

$\frac{du}{dx} = 3x^2 \frac{dv}{dx} = -e^{-x}$

$= 3x^2 e^{-x} - x^3 e^{-x}$

6) $f(x) = (3x^2 + 2x + 1)^5$

$u = 3x^2 + 2x + 1$

$f(x) = u^5$

$\frac{du}{dx} = \frac{d}{dx}(u^5) = 5u^4 \cdot u'$

$u' = 6x + 2$

$f'(x) = 5(3x^2 + 2x + 1)^4 (6x + 2)$

7) $f(x) = (\sin x)^4$

let $u = \sin x$

$f(x) = u^4$

$\frac{du}{dx} = \frac{d}{dx}(u^4) = 4u^3 \cdot u'$

$u' = \cos x$

$f'(x) = 4\sin^3(x) \cos x$

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8) $f(x) = e^{(3x^2+2x)}$

let $v = 3x^2+2x$

$f(x) = e^v$

$\frac{dv}{dx} = \frac{d}{dx}(e^v) = e^v \cdot v'$

$v' = 6x+2$

$f'(x) = e^{3x^2+2}(6x+2)$

9) $f(x) = \ln(5x^2+3x)$

let $v = 5x^2+3x$

$f(x) = \ln v$

$\frac{dv}{dx} = \frac{d}{dx}(\ln v) = \frac{v'}{v}$

$v' = 10x+3$

$f'(x) = \frac{10x+3}{5x^2+3x}$

10 $f(x) = \sqrt{x^3+4x+1}$

$f(x) = (x^3+4x+1)^{1/2}$

let $v = x^3+4x+1$

$f(x) = v^{1/2}$

$f'(x) = \frac{1}{2} v^{-1/2} v'$

$v' = 3x^2+4$

$f'(x) = \frac{1}{2}(x^3+4x+1)^{-1/2}(3x^2+4)$

$f'(x) = \frac{3x^2+4}{2\sqrt{x^3+4x+1}}$

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Problem 2,

$g(1) = x^2 - 3x + 2$

1) derivative of $f(x) \cdot g(x)$ at $x = 1$

$g(1) = 1^2 - 3 \times 1 + 2 = 0$

$g'(x) = 2x - 3$

using chain rule: $(fg)'(1) = f'(1)g(1) + f(1)g'(1)$

$= 5 \times 0 + 5(-1)$

$= -5$

$f(0) = 3$, $f'(0) = -1$

$g'(0) = 2 \times 0 - 3 = -3$

2) Derivative of $\frac{f(1)}{g(1)}$ at $x = 0$

$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$

$\left(\frac{f}{g}\right)'(0) = \frac{(-1) \times 2 - 3 \times (-3)}{2^2} = \frac{7}{4}$

3) Derivative of $f(g(x))$ at $x = 0$

$[f(g(x))]' = f'(g(x)) \cdot g'(x)$

$g(0) = 2$

$f'(2) = 3$ $\quad g'(x) = 2x - 3$

$g'(0) = -3$

Substituting, $[f(g(x))]'x = 0 = f'(2) \cdot (-3)$

$= 3(-3)$

$= -9$

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4.2.1) $g(x) = x^2 - 4x + 5$ at interval $[1, 4)$

$g'(x) = 2x - 4$

$g'(1) = 2 \times 1 - 4 = -2$

$g(1) = 1 - 4 + 5 = 2$

$g(4) = 16 - 16 + 5 = 5$

min at $g(1)$ and max at $g(4)$

2) $h(x) = \frac{x^2}{2} - 2x$ at interval $[-2, 3)$

$h'(x) = x - 2$

$\therefore x = 2$, is critical point

$h(-2) = \frac{4}{2} + 4 = \frac{4+8}{2} = \frac{12^6}{2} = 6$

$h(3) = \frac{9}{2} - 6 = \frac{9-12}{2} = \frac{-3}{2}$

$h(2) = \frac{4}{2} - 4 = \frac{4-8}{2} = \frac{-4^2}{2} = -2$

3) $p(x) = -3x^3 + 9x^2 + 7$ at interval $[-1, 2]$

$p'(x) = 9x^2 - 18x$

$p(-1) = -3 - 9 + 7 = -5$

$p(2) = 3 \times 8 - 9 \times 4 + 7 = 24 - 36 + 7 = -5$

$p(0) = 7$

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4) $f(x) = x^2 - 4x + 6$ at interval $[0, 3]$

$f'(x) = 2x - 4$

$f(0) = 0 - 0 + 6 = 6$

$f(3) = 9 - 12 + 6 = 3$

min at $f(3)$ and max at $f(0)$

5) $q(x) = -2x^3 + 3x^2 + 12x$ at interval $[-1, 2]$

$q'(x) = -6x^2 + 6x + 12$

$q(-1) = 2 + 3 - 12 = 7$

$q(2) = -16 + 12 + 24 = -16 + 36 = 20$

4.3.3) evaluate gradient function of given points

1) $f(x, y) = x^2 + y^2$ at

a) $(0, 0) = (2 \times 0, 2 \times 0) = (0, 0)$

b) $(1, 3) = (2 \times 1, 2 \times 3) = (2, 6)$

c) $(-1, -5) = (2 \times -1, 2 \times -5) = (-2, -10)$

2) $f(x, y, z) = x^3 z - 2y^2 + x + 5z$ at:

a) $\frac{\partial f}{\partial x} = 3x^2 z - 2y^2 \quad \frac{\partial f}{\partial y} = -4yx \quad \frac{\partial f}{\partial z} = x^3 + 5$

$\nabla f = (3x^2 z - 2y^2, -4yx, x^3 + 5)$

a) $(1, 1, -4) = (3 \times 1 \times (-4) - 2 \times 1, -4 \times 1 \times 1, 1 + 5) = (-14, -4, 6$

b) $(0, 1, 0) = (3 \times 0 \times 0 - 2 \times 1, -4 \times [0 \times 0, 0 + 5, 0 + 5) = (-2, 0, 5$

c) $(-3, -2, 1) = (3 \times 9 \times 1 - 2 \times 4, -4 \times (-2) \times (-3), (-3)^3 + 5)$

$= (19, -24,$

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3) $m(x,y) = \ln(x^2+y^2+1)$ at:

$\frac{\partial m}{\partial y} = \frac{2x}{x^2+y^2+1}$

$\frac{\partial m}{\partial y} = \frac{2y}{x^2+y^2+1}$

$Dm = \left(\frac{2x}{x^2+y^2+1}, \frac{2y}{x^2+y^2+1}\right)$

a) $(0,0) = \left(\frac{2\times0}{0+0+1}, \frac{2\times0}{0+0+1}\right) = (0,0)$

b) $(1,-1) = \left(\frac{2\times1}{1^2+(-1)^2+1}, \frac{2\times(-1)}{1+1+1}\right) = \left(\frac{2}{3}, -\frac{2}{3}\right)$

c) $(-2,3) = \left(\frac{2\times(-2)}{4+9+1}, \frac{2\times3}{4+9+1}\right) = \left(\frac{-2}{7}, \frac{3}{7}\right)$

2) maxima & minima of multivariate function

1) $m(x,y) = -x^2-y^2+4x+6y$

$\frac{\partial m}{\partial x} = -2x+4=0, \quad x=2$

$\frac{\partial m}{\partial y} = -2y+6=0, \quad y=3$

$m(2,3) = -4-9+8+18 = 13$

$m(0,0)=0, \; m(4,0)=0, \; m(4,5)=5, \; m(4,5)=5$
$m(2,0)=4, \; m(2,5)=5, \; m(0,3)=9, \; m(4,3)=9$

∴ max at $(2,3)$ & min at $(0,0)$ and $(4,0)$

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2) $g(x,y) = xy + x^2 - 4y + 6$

$\frac{\partial g}{\partial x} = y + 2x = 0$

$\frac{\partial g}{\partial y} = x - 4 = 0;\ x = 4$

$g(-2,3) = -6 + 4 - 12 + 6 = 8$

$g(2,1) = 2 + 4 - 4 + 6 = 8$

$g(2,3) = 6 + 4 - 12 + 6 = 4$

∴ on edge, $x = -2$, $g = -2y + 4 - 4y + 16 = 10 - 6y$, max at $y = 1$; $g = 4$. min at $y = 3$; $g = 8$

$g(-2,4) = -8 + 64 - 16 + 6 = 46$

3) $p(x,y) = x^3 - 3xy + 4$ interval $x: [0,2]$ and $y [0,3$

$\frac{\partial p}{\partial x} = 3x^2 - 3y = 0,\ y = x^2$

$\frac{\partial p}{\partial y} = -3x = 0 \therefore x = 0$, then $y = 0$

$p(0,0) = 4$

$p(2,0) = 8 + 4 = 12$

$p(0,3) = 4$

$p(2,3) = 8 - 18 + 4 = -6$

∴ max at $p(2,0)$ and min at $p(2,3)$

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4) $h(x,y) = 2x^2 + 3y^2 - 12x - 18y + 20$

$\frac{\partial h}{\partial x} = 4x - 12 = 0, \quad x = 3$

$\frac{\partial h}{\partial y} = 6y - 18 = 0, \quad y = 3$

$h(3,3) = 18 + 27 - 36 - 54 + 10 = -25$

$h(1,0) = 2 + 0 - 12 - 0 + 20 = 10$

$h(1,4) = 2 \times 1 + 3 \times 16 - 12 \times 1 - 18 \times 4 + 20 = -14$

$h(5,4) = 2 \times 25 + 3 \times 16 - 12 \times 5 - 18 \times 4 + 20 = -14$

$\therefore$ max at $(1,0)$ and $(5,0)$, min at $(3,3)$

5) $f(x,y) = x^2 + y^2 - 4x - 6y + 13$

$\frac{\partial f}{\partial x} = 2x - 4 = 0, \quad x = 2$

$\frac{\partial f}{\partial y} = 2y - 6 = 0, \quad y = 3$

$\therefore f(2,3) = 4 + 9 - 8 - 18 + 13 = 0$

$f(1,1) = 1 + 1 - 4 - 6 + 13 = 5$

$f(3,1) = 9 + 1 - 12 - 6 + 13 = 5$

$f(1,4) = 1 + 16 - 4 - 24 + 13 = 2$

$f(3,4) = 9 + 16 - 12 - 24 + 13 = 2$

$\therefore$ max at $f(1,1)$ and $f(3,1)$

$\therefore$ min at $(2,3)$ //

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